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Review

Review

Recreational substance use among international travellers

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Abstract

Background: Drug tourism reflects the expanding illicit drug market, posing health risks in unfamiliar travel settings. Existing knowledge specifically addressing substance use among international travellers is sparse and has not been reviewed to date. This review aimed to describe the recreational substance abuse in international travellers.

Methods: A literature search was conducted on PubMed, Google Scholar and Scopus using keywords related to recreational substances and international travellers. A total of 11 021 articles were reviewed, charted and summarized for the evidence on prevalence, patterns and characteristics of substance abuse and their health- and non-health-related problems on international travellers.

Results: A total of 58 articles were included. Most were cross-sectional studies and review articles. In total, 20 articles addressed the prevalence of substance abuse in travellers, 33 looked at characteristics and patterns of substance abuse in travellers and 39 investigated the health- and non-health-related problems from substance abuse. Estimated prevalence of recreational substances abuse varied from 0.7% to 55.0%. Rates of substances abuse were 9.45–34.5% for cannabis, 20.4–35.9% for alcohol intoxication, 2.82–40.5% for MDMA, 2–22.2% for cocaine, 2–15% for psychedelic agents and 2% for methamphetamine. The prevalence varied according to travellers' characteristics and travel destinations. Direct health problems included neuropsychiatric problems. Indirect problems included accident and unintentional injuries, crime and violence, risky sexual behaviours and sexual violence and blood-borne infections. Non-health-related problems included air rage, deportation and violation of local laws.

Conclusion: Substance abuse among international travellers is an underestimated problem that requires intervention. These findings emphasize the importance of addressing this issue to mitigate both health and well-being problems among travellers whilst promoting safer and more responsible travel experiences. In the context of travel health practices, practitioners should counsel travellers whose itineraries may include substance abuse, informing them about associated risks and consequences.

Key words: Illicit drug, health impact, prevalence, contributing factors, international traveller, drug legislation

Introduction

Substance abuse is an ongoing problem globally. Despite its recognized dangers, substance use continues to proliferate, particularly in low- and middle-income countries. [1] As the number of drug users grows, it is conceivable that a substantial number of these individuals may be potential drug tourists.[2]

'Drug tourism' or 'narcotourism' can be defined as a travelling by which an individual is attracted to a particular destination due to accessibility of substances and related services.[3,4] A 'drug tourist' is someone using drugs whilst exploring different destinations, with drug use can be either the primary purpose of travel or complimentary factor.[4,5]

Substances can be categorized as 'hard drugs' and 'soft drugs'. Hard drugs, including opioids, anxiolytics/hypnotics, amphetamines, cocaine, hallucinogens, volatiles and gamma-hydroxybutyric acid (GHB), have substantial physical and mental health impacts, often leading to addiction and potential fatality.[6] In contrast, soft drugs, like alcohol, nicotine and cannabis, are less harmful, yet unsafe for excessive use.[6]

Substance abuse during travel can lead to several risks that may be unrealised by unwary travellers. These risks include incarceration; being threatened by local criminals; violence and accidents; and health problems such as neuropsychiatric problems, exposure to blood-borne diseases and risky sexual practices.[3,7,8] However, the risks of substance abuse in foreign destinations are rarely discussed during pre-travel consultations.

Current knowledge regarding substance abuse among international travellers is sparse which complicates the provision of pre-travel advice. The purpose of this review was to synthesize the current evidence and identify knowledge gaps about common recreational substances used by international travellers, prevalence, pattern of abuse and impacts on health and well-being of travellers.

Methods

The protocol for this review was registered with the Open Science Framework (https://osf.io/5pncd). We adhered to established methodological guidance for the conduct of scoping reviews and report this review in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Checklist (Supplementary File 1).

Inclusion and exclusion criteria for studies selection

The inclusion criteria were: (i) Articles from peer-reviewed academic journals or pre-print servers; (ii) any studies, including quantitative studies, qualitative studies, review, case reports, perspectives and letters and (iii) studies that related to substances abuse, use of prescribed medications and bringing substances across international borders in travellers. Excluded articles were: (i) not related to recreational drugs; (ii) not travel-related; (iii) not related to prevalence, pattern of substance abuse and the impact of substances in international travellers; (iv) incorrect publication type; (v) non-human studies and (vi) foreign language.

Data sources and search strategy

The peer-reviewed literature was accessed through electronic indexing databases including PubMed, Google Scholar and Scopus. Combinations of keywords were used which related to recreational substance and international travellers. The keywords and search algorithms were available in Supplementary File 2.

Data extraction

Following the removal of duplicates, titles and abstracts underwent dual screening, with any ambiguities resolved through discussion between the two reviewers or, when needed, within the broader review team. For studies identified as potentially relevant or unclear during the title/abstract screening, two reviewers independently examined the full text. Discrepancies were addressed through dialogue between the two reviewers, and any uncertainties were deliberated with a third reviewer and/or the larger review team. At this point, a conclusive determination on inclusion was reached, and rationales for exclusion were recorded. We used the web-based application Rayyan (https://rayyan.qcri.org/) to manage the collection and de-duplication of records.

A reviewer extracted and charted study characteristics and data regarding the international travellers which encompassed tourists, backpackers, international students, expatriates and drug traffickers; recreational substances which were alcohol, cannabis, cocaine, MDMA (ecstasy and molly), methamphetamine and amphetamine, psychedelic agents such as LSD and other related drugs such as ketamine and heroin; patterns and characteristics of substance abuse by international travellers; health problems from substance abuse in international travellers and non-health-related problems due to recreational substance abuse in international travellers. Findings were extracted as reported within the study papers themselves. One reviewer reviewed all extracted data.

Results

Search results and characteristics of included studies

The PRISMA-ScR diagram is shown in Figure 1. A total of 11 021 articles were initially retrieved from three databases. Through title and abstract screening, duplicated and unrelated studies were excluded, leaving 515 potentially relevant articles for further detailed assessment. Subsequently, 458 articles were excluded, of which 364 were unrelated to recreational substances, 63 were not associated with international travel, 13 were in foreign languages, nine were not related to prevalence, patterns of substance abuse, and the impact of substances, seven were of incorrect publication types and two were non-human studies. Finally, 58 articles were included in the scoping review.

The characteristics and outcomes of included articles are presented in Table 1. Among them, 23 were cross-sectional studies, 16 were reviews and seven were qualitative studies, whilst the remainder encompassed case–control studies, cohort studies, interventional studies, case series, perspectives and letters. The sample sizes of the studies varied, ranging from 8

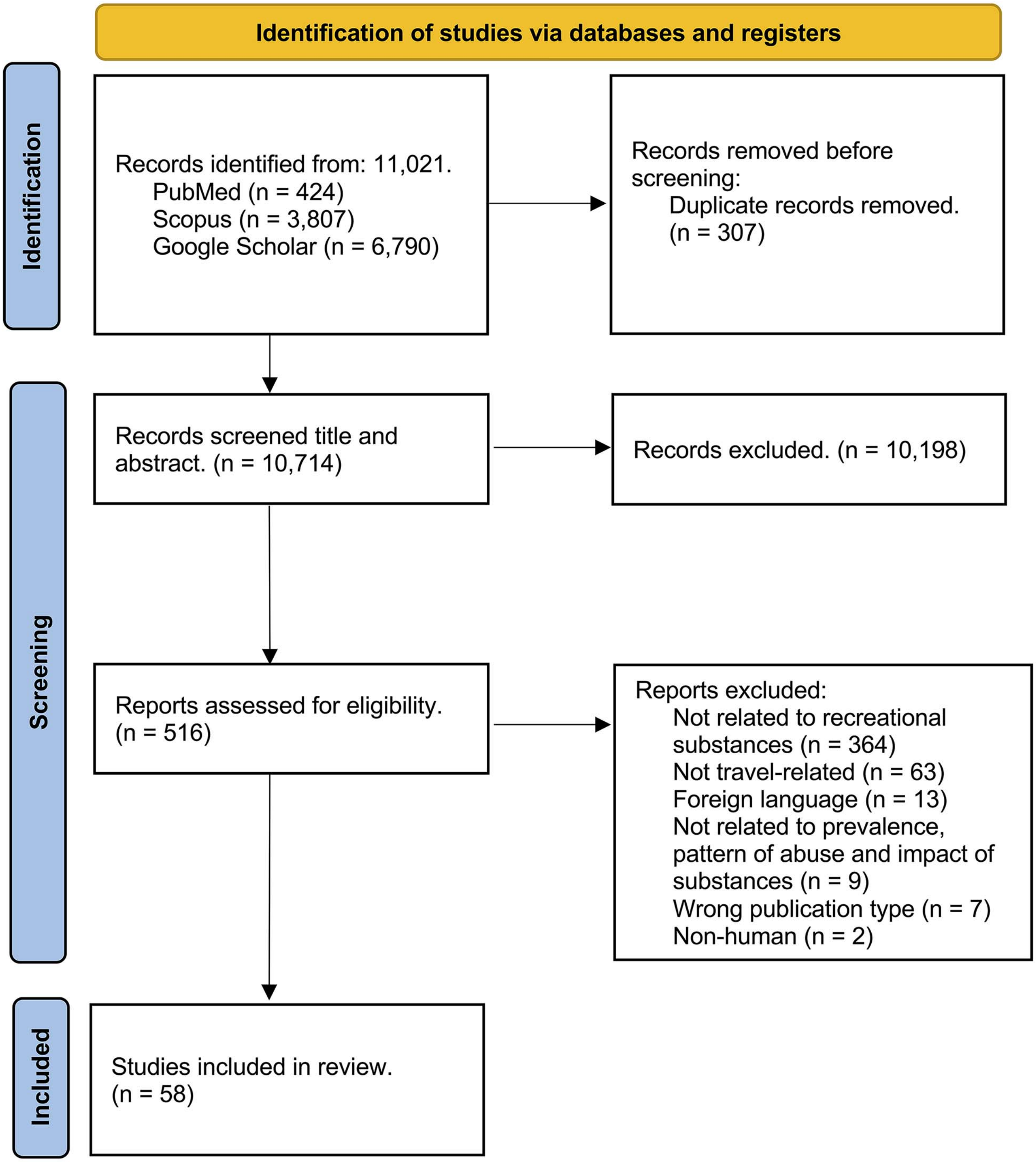

Figure 1 PRISMA-ScR diagram

to 6502. Twenty articles specifically addressed the prevalence of substance abuse in travellers, whilst characteristics and patterns of substance abuse in travellers were discussed in 33 articles. Furthermore, 39 studies included health- and non-health-related problems associated with substance abuse.

Prevalence of substance abuse among international travellers

The prevalence of recreational substances abuse among travellers can vary significantly, ranging from 0.7% to 55%.[7,9–20] The most frequently used substances include cannabis, alcohol, cocaine

Table 1 Characteristics of included studies

Author, year	Study design	Duration of study	Type of substances studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
Lange, 1989[8]	Review	N/A	Alcohol, prescribed medications (benzodiazepines, barbiturates, sedatives, stimulants) and recreational drugs (cocaine, opioids, heroin), methadone	Worldwide	International travellers (n = N/A)	N/A	N/A	Complications of the physiologic adjustments, deterioration and precipitation of clinical conditions during travel, accidents and legal difficulties whilst travelling and exposure to serious infectious diseases worldwide.
Allen, 1992[26]	Qualitative	1989 to 1990	Psychoactive mushroom	Koh Samui and Koh Pha-ngan, Thailand	International travellers (n = N/A)	N/A	Psychoactive mushrooms are cultivated and sold directly to tourists or to resort restaurants for mixing in food items including omelettes and soups, which sometimes, adulterated with more potent artificial hallucinogens such as LSD.	N/A
Young, 1995[12]	Case–control	Aug 1990 to Jan 1991	N/A	Honolulu, USA	International travellers ($n = 30$)	30%	N/A	Psychiatric morbidity including mania and depression.[b]
Potasman, 2000[10]	Cross-sectional	Jan 1994 to June 1996	Recreational substances, including cannabis, ecstasy, amphetamines, LSD and mushrooms.	Worldwide	International travellers ($n = 117$)	22.2%	N/A	Sleeping disturbances, fatigue and dizziness
Bellis, 2000[13]	Cross-sectional	Jul to Sep 1999	Amphetamine, ketamine, cannabis, ecstasy, LSD, cocaine and GHB	Ibiza, Spain	International travellers ($n = 846$)	0.7–33.1%	Individuals using most substances fell whilst travelling compared to use at home. However, when used, the frequency of use for all substances was higher in Ibiza than at home.	Unprotected sex and multiple sex partners. Unspecified cause of hospital visits.
Beny, 2001[41]	Case series	N/A	Cannabis, LSD and ecstasy	Worldwide	International travellers ($n = 15$)	N/A	N/A	Eight of 15 travellers who had neuropsychiatric problems had history of recreational drug use, which results in depressive disorder, panic, anxiety, acute psychosis and agoraphobia.
Bellis, 2002[59]	Review	N/A	Cannabis, ecstasy, amphetamine and cocaine	N/A	International travellers (n = N/A)	N/A	N/A	Clubbing aboard increased the risk to the health hazards due to unfamiliarity of the destination country. Geographical and language barriers complicate the access to help, health services and preventive measures. Unfamiliarity with local legislatures can put travellers under judicial services. Lack of restrictions at home may put travellers at an increased exposure to substances.

(*Continued*)

Table 1 Continued

Author, year	Study design	Duration of study	Type of substances studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
McInnes, 2002[45]	Review	N/A	Alcohol	N/A	International travellers (n = N/A)	N/A	N/A	Alcohol was the risk factor of fatal injury and the primary cause of vehicle crashes in tourists.
Paz, 2004[21]	Intervention	August to Oct 1999	Cannabis, ecstasy, LSD, mushrooms, cocaine, opium and cactuses	Southeast Asia and South America	International travellers ($n = 223$)	36.8%	Most travellers used drugs in Southeast Asia. Female travellers travelling to Southeast Asia, education ≤ 12 school years, age ≤ 25 years and lack of malaria chemoprophylaxis contributed to drug abuse.	N/A
Hughes, 2004[17]	Case–control	2002	Alcohol, amphetamines, ketamine, cannabis, ecstasy, LSD, cocaine, GHB	Ibiza, Spain	Occupational travellers ($n = 92$)	23.9–91.3%	Occupational travellers had longer median stay duration in Ibiza, used more types of substances and took more tablets of ecstasy than regular travellers.	N/A
Richens, 2005[52]	Review	N/A	Alcohol and unspecified recreational drugs	Worldwide	International travellers (n = N/A)	N/A	N/A	Alcohol and recreational drugs were among the risk factors of having sex whilst travelling. High level of alcohol consumption in travellers can lead to sexual violence, particularly in lone female travellers.
Segev, 2005[11]	Cross-sectional	Feb 2002 to Nov 2003	Cannabis, mushrooms, LSD, ecstasy, cocaine, amphetamines, opium, heroin and MDMA	Southeast Asia	International travellers ($n = 430$)	54.3%	Males abused drugs more than females. People with academic degrees abused less than others. Prior drug use, Israeli nationality, travelling to India, smoking and travel alone are predictors of drug abuse	N/A
Darrow, 2005[18]	Cross-sectional	Feb to March 2004	Alcohol, cannabis, nitrite inhalant, cocaine, ecstasy, methamphetamine, ketamine and GHB	Miami, USA	Local and international travellers (MSM) (Total $n = 407$[c]	32%[c]	N/A	Tourists were more likely than locals to use cocaine and ketamine and engaging in unprotected receptive anal intercourse
Uriely, 2005[36]	Qualitative	2000–2002	Cannabis, ayahuasca, peyote and *San Pedro* cactus	Worldwide	International travellers (n = N/A)	N/A	Drug-related tourists' experiences involved a pursuit of pleasure or a quest for profound and meaningful experiences.	N/A

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Table 1 Continued

Author, year	Study design	Duration of study	Type of substance s studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
Streeton, 2006[54]	Cross-sectional	2001–2003	Injections, including injected drug	Worldwide	International travellers ($n = 503$)	N/A	N/A	Four percent of travellers travelling to hepatitis B endemic areas had injections which were risk of potential exposure to hepatitis B.
Benotsch, 2006[19]	Cross-sectional	Sep 2004	Alcohol, cannabis, cocaine, ecstasy, LSD, methamphetamines, poppers, ketamine, rophynol and GHB	Florida, USA	Local and international travellers (MSM) (Total $n = 304$, international travellers $n = 10$)	16–56%[c]	N/A	Individuals using substances reported higher rates of sexual risk behaviours, including unprotected anal sex and asking about partner's HIV status.
Belhassen, 2007[35]	Qualitative	2000–2002	Cannabis	Netherlands, Egypt and India	International travellers ($n = 18$)	N/A	Motives of cannabis use included experimentation, loosening social control, recreation, leisure, fun-seeking, search for cannabis authenticity, self-identity and part of smuggling cannabis across border	N/A
Bellis, 2007[15]	Cross-sectional	Apr to Nov 2005	Cannabis, ecstasy, amphetamine, methamphetamine, cocaine, LSD ketamine, GHB, heroin and steroids.	Sydney and Cairns, Australia	Backpackers ($n = 1008$)	44.5–55.0%	Backpackers used substances more frequently whilst travelling than at home. Regular club goers, male, stayed in Sydney, travelling alone, travelling >4 weeks and daily smoking and drinking are among the risks of recreational drug use whilst backpacking.	N/A
Benotsch, 2007[51]	Cross-sectional	2005	Alcohol and recreational drugs	New Orleans, USA	Local and international travellers (MSM) (Total $n = 132$, international travellers $n = 8$)	N/A	N/A	Excessive alcohol use was related to unprotected anal sex. Substance use with sexual activity was associated with higher risky sexual behaviours. Negative correlation between using drugs before anal sex and the disclosure of HIV status to partners was observed.

(*Continued*)

Table 1 Continued

Author, year	Study design	Duration of study	Type of substances studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
Guzman, 2008[24]	Review	N/A	Hallucinogenic mushrooms	Mexico	International travellers (n = N/A)	N/A	Sacred mushroom ceremony has become more commercialized for tourism purpose in recent years. Tourists may be initially offered only one or two pairs of mushroom, but can pay more for additional dose.	Mental disorders if taken in excessive doses (>12 mushrooms)
Hughes, 2008[16]	Cross-sectional	Jul to Aug 2007	Alcohol, cannabis, ecstasy, cocaine, amphetamines, ketamine and GHB	Ibiza and Majorca, Spain	International travellers (*n* = 3003)	0.9–22.4%	Drug users that involved in violence tended to prefer venues that sell cheap drinks, offered opportunities for sex, loud music, have games and where people got drunk.	Violence during holiday was associated with alcohol drunkenness and cannabis or cocaine use during holiday.
Ilhan, 2009[37]	Cross-sectional	N/A	Cannabis, ecstasy, heroin, cocaine and solvents	Turkiÿe	International university students (*n* = 1720)	3.3–12.8%[d]	Students whose families reside aboard increased the risk of substance use, partly due to peer effects and reduced parental control.	N/A
Nielsen, 2009[55]	Cross-sectional	Jun to Jul 2007	Injections, including injected drug	Worldwide	International travellers (*n* = 1090)	N/A	N/A	Injections accounted for 17% of risky activities for acquiring hepatitis B in travellers.
Bellis, 2009[14]	Cross-sectional	Jul to Aug 2007	Alcohol, cannabis, ecstasy, cocaine, amphetamines, ketamine and GHB	Ibiza and Majorca, Spain	International travellers (*n* = 3003)	0.96–22.43%	Drug use whilst travelling was associated with male gender, substance use at home, travel to Ibiza, Spanish nationality and having drunkenness at least one time per week. First-time ecstasy user was found in 0.8–8.6% of travellers. Relapsing into using recreational drugs ranged between 1.3 and 6.8%.	N/A
Vivancos, 2010[33]	Prospective cohort	Jun to Nov 2006	Alcohol, amphetamines, cannabis, cocaine and ecstasy	Worldwide	International travellers (*n* = 264)	10.6%	Alcohol and drug use are more frequent in students who travelled during summer break.	Drinking excessively is associated with new sexual partners during travelling aboard.
Hughes, 2011[7]	Cross-sectional	Jul to Aug 2009	Cannabis, ecstasy, cocaine, amphetamines, ketamine, GHB, alcohol	Cyprus, Greece, Italy, Portugal, Spain	International travellers (*n* = 6502)	10.9–35.9%	Most common used substance was cannabis. Use of drugs on holiday was highest among travellers to Cyprus and German travellers to Portugal	Violence and unintentional injury is associated with alcohol consumption, drunkenness and use of recreational drugs.

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Table 1 Continued

Author, year	Study design	Duration of study	Type of substances studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
Cabada, 2011[22]	Cross-sectional	Oct to Dec 2004	Alcohol	Cusco, Peru	International travellers ($n = 1120$)	20.4%	Travellers who are male, single, younger than 26 years old and travelling alone or with friends are more likely to have alcohol intoxication.	Alcohol intoxication is associated with unsafe food choices, recreational drug use and risky sexual activity.
Klunge de-Luze, 2014[9]	Cross-sectional	Jan 2006 to Dec 2008	Cannabis, ecstasy, cocaine, LSD, NMDA, heroin, amphetamines, hallucinogenic mushrooms, alcohol	Worldwide	International travellers ($n = 3537$)	5–14%	At-risk alcohol and consumption and substance use at home, smoking and leisure trip contributes to substance use during travel	N/A
Kelly, 2014[34]	Cross-sectional	Jul to Aug, 2009	Alcohol, cannabis, ecstasy, amphetamines, ketamine and GHB	Ibiza, Spain	Occupational travellers ($n = 171$)	85.3%	Most of workers who consumed alcohol reported to have getting drunk at least once a week. Most of casual workers on Ibiza had used drugs. Most common used drugs were ecstasy, cocaine, ketamine, cannabis, amphetamines and GHB. Most workers in Ibiza reported using drugs they never used at home and the frequency of use is increased whilst in Ibiza.	Unprotected sex and multiple sexual partners are frequently under the influence of alcohol.
Cappeletti, 2015[56]	Systematic review	N/A	Cocaine, heroin, opioids, methamphetamine and cannabis	N/A	Drug traffickers (n = N/A)	N/A	Drug traffickers usually wrapped drugs in package and swallowed or placed in various anatomical cavities or orifices.	Serious health consequences were chemical and mechanical complications. Chemical complications included the systemic absorption of drugs or intoxication due to package leakage. Mechanical complications included gastrointestinal obstruction.
Pereira, 2016[4]	Qualitative	N/A	Cannabis	Netherlands	International travellers (n = N/A)	N/A	Coffee shops which offered cannabis and cannabis-related products and services in Amsterdam attracts up to 16% of tourists visiting the Netherlands. Cannabis tourism was an important part of tourism industry in the Netherlands.	N/A

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Table 1 Continued

Author, year	Study design	Duration of study	Type of substances studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
Aresi, 2016[38]	Systematic review	N/A	Recreational substances including alcohol and cocaine	N/A	International university students (n = N/A)	N/A	Students in short-term or semester programme aboard increased alcohol. Full bachelor's or master's degree international students consumed less substances than domestic students. Several factors, such as perceptions of peer drinking norms and adjustment aboard, affects the at-risk alcohol consumption.	N/A
Crawford, 2016[53]	Systematic review	N/A	Alcohol and recreational drugs, including cannabis	Worldwide	International travellers and expatriates (n = N/A)	N/A	N/A	Risk factors of HIV, blood-borne virus and STDs included using alcohol and drugs, among others. Alcohol reduced inhibition and judgement, which facilitated unsafe sex.
Stewart, 2016[44]	Review	N/A	Alcohol	N/A	International travellers (n = N/A)	N/A	N/A	Alcohol use is one of the major risks of traffic accidents, drowning and fall in travellers to low- and middle-income countries.
Prayag, 2016[23]	Qualitative	2011	Ayahuasca	Iquitos, Peru	International travellers ($n = 6$)	N/A	Main motives for experiencing ayahuasca included purging individual's self of modern lifestyle, spiritual experience, self-discovery, learning the nature and health purposes.	N/A
Peden, 2016[47]	Cross-sectional	Jul 2002 to Jun 2012	Alcohol	Australia	International travellers ($n = 123$)	12.2%	N/A	Alcohol contributed to fatal drowning in travellers. Around 69% of fatal drowning cases due to alcohol had blood-alcohol content (BAC) greater than upper legal limit for operating motor vehicle.
Flaherty, 2017[3]	Perspectives	N/A	Cannabis, cocaine, ecstasy, amphetamines, GHB, psychedelic mushrooms, hallucinogenic peyote cactus, ayahuasca, opioids and alcohol	Worldwide	International travellers, occupational travellers and drug traffickers (n = N/A)	N/A	Male gender, regular club goers, travelling alone, long-term travelling >4 weeks and drinking alcohol or smoking for five or more days a week contributed to substance use. Shamanic rituals in South America attracted spiritual travellers. Occupational travellers to Balearic Islands have high rates of substance abuse. Full moon parties in India are popular among drug tourists	Accidents, violence, unprotected sex, sexual violence, risk of STDs, drug-induced psychosis, acute intoxication, subjected to local laws

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Table 1 Continued

Author, year	Study design	Duration of study	Type of substances studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
Lu, 2018[31]	Letter	N/A	Alcohol and recreational drugs, such as LSD, MDMA, ketamine, synthetic cathinones and phenethylamines.	Asia, Europe and USA	International travellers (n = N/A)	N/A	Music festivals attract young travellers, some of them will try recreational drugs.	Aggressive beahviour. Drug intoxication which lead to hospitalization and death.
Bauer, 2018[43]	Letter	N/A	Ayahuasca	Peru, Colombia and Ecuador	International travellers ($n = 10$)	N/A	N/A	All fatalities are possibly attributed to ayahuasca use alone or mixing with other substances which can cause the serotonin syndrome.
Svensson, 2018[49]	Meta-analysis and systematic review	N/A	Alcohol and recreational drugs including cannabis	Worldwide	International travellers and backpackers (n = N/A)	N/A	N/A	Alcohol and drug uses are associated with casual sex and non-condom use whilst travelling.
Ying, 2019[29]	Review	N/A	Cannabis	Worldwide	International travellers (n = N/A)	N/A	Tourists may perceive that leisure travel allows them to engage in risky activities including cannabis tourism. Travelling aboard can allows travellers to use cannabis which is prohibited or immoral at their home.	N/A
McLintion, 2020[58]	Systematic review	N/A	Alcohol, prescribed medications and unspecified drugs	Worldwide	Internatuional travellers (n = N/A)	N/A	N/A	Alcohol consumption, use of prescribed medications, such as benzodiazepine and the use of recreational or therapeutic drugs rank among the underlying causes of air rage
Pereira, 2020[2]	Review	N/A	Cannabis, heroin, cocaine, ecstasy, LSD, hallucinogenic mushroom, opium, methamphetamine, *San Pedro* cactus, ayahuasca	Worldwide	International travellers (n = N/A)	N/A	Changes in drug policies in several countries could play a role in shaping the drug tourism. Each destinations had their own characteristics of drug tourism, such as drug-friendly zones in European cities and cocaine and hallucinogenic agents use patterns in South America.	N/A
Bonny-Noach, 2020[27]	Perspectives	N/A	Cannabis	Worldwide	International travellers (n = N/A)	N/A	Cannabis tourism is growing worldwide. Use of cannabis in travel destinations known for drug tourism is tolerated, legal or decriminalized.	Arising mental illness whilst travelling aboard which may need medical evacuation.

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Table 1 Continued

Author, year	Study design	Duration of study	Type of substances studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
Evers, 2020[20]	Cross-sectional	2017 to 2018	Amphetamines, ecstasy, GHB, ketamine and cocaine	Worldwide	International travellers (MSM) ($n = 405$)	42.7%	Almost 28% of MSM travellers to Western countries reported chemsex. In contrast, only 1% of travellers to Non-Western countries had chemsex.	In MSM travellers who had chemsex whilst travelling in Western countries, 26.7% was diagnosed with STD.
Firth, 2020[39]	Cross-sectional	2018	Cannabis	Australia, China, Costa Rica, France, Germany, Ireland, Italy, Japan, Mexico, South Africa, Spain, United Kingdom	International university students ($n = 2086$)	0.0–35.9%	Students who were male, had history of pre-departure cannabis, cigarettes and e-cigarette use and longer duration of study aboard had more frequency of cannabis use. Studying in Germany and Spain was likely to use cannabis.	N/A
Houle, 2021[25]	Scoping review	N/A	Ayahuasca	N/A	International travellers (n = N/A)	N/A	Some travellers use ayahuasca for the treatment of depression and grief.	Psychiatric events such as hallucinations, agitation, paranoia, delusion, hyperactivity, disorientation and aggression. Ayahuasca can cause serotonin syndrome. Non-psychiatric events such as vomiting, loss of balance, seizure, tremor, sweating, tachycardia, hypertension, mydriasis, hyponatremia, priapism and rhabdomyolysis. Five fatalities were reported.
Coyle, 2021[57]	Cross-sectional	2000–2020	Alcohol and unspecified drugs	Worldwide	International travellers ($n = 270$)[b]	N/A	N/A	The most common precipitator of air-rage was alcohol consumption.
Sapsirisavat, 2021[46]	Cross-sectional	Jan 2015 to Dec 2019	Alcohol and recreational drugs (unspecified)	Thailand	International travellers ($n = 720$)	13.2%	N/A	Alcohol drinking was associated with hospitalization due to traffic injuries.
Bonny-Noach, 2022[28]	Perspective	N/A	Cannabis	Worldwide	International travellers (n = N/A)	N/A	The cannabis policy changes to attack tourists after pandemic.	Changing drug policy might affect health problems
Charoenwisedsil, 2023[61]	Letter	N/A	Cannabis	Thailand	International travellers (n = N/A)	N/A	N/A	Psychological effects and accidents. Travellers arrested for carrying marijuana from Thailand to other countries.
Potin, 2023[42]	Review	N/A	Alcohol, cannabis, cocaine, amphetamines and prescribed medications including benzodiazepine	N/A	International travellers (n = N/A)	N/A	N/A	Several prescribed medications, such as benzodiazepine and recreational substances can aggravate psychiatric symptoms which may result in emergency visits during travelling.

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Table 1 Continued

Author, year	Study design	Duration of study	Type of substances studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
Kissane, 2023[60]	Review	N/A	Prescribed psychotropic drugs and narcotic drugs including cannabis	Worldwide	International travellers using prescribed psychotropic medications (n = N/A)	N/A	N/A	Most countries use International Narcotics Control Board template to communicate about the restrictions on psychotropic and narcotic drugs importation. Most countries require valid medical prescription. There were considerable variety between the prohibited substances and amount of maximum imported quantities that can confuse travellers.
Wilcox-Pidgeon, 2023[48]	Cross-sectional	2008–2018	Alcohol, prescribed medications and unspecified recreational drugs	Australia	International travellers, international students, work-related travellers and working holiday makers ($n = 201$)	N/A	N/A	Blood alcohol concentration ≥0.05%, prescribed medications and recreational drugs was recorded in 10.9%, 17.4% and 6.0% of fatal drownings, respectively.
Pedersen, 2023[50]	Cross-sectional	2017–2019	Alcohol and unspecified recreational substances	Australia, China, Costa Rica, France, Germany, Ireland, Italy, Japan, Mexico, South Africa, Spain and UK	International college students ($n = 2428$)	N/A	N/A	Heavy alcohol drinking had the greatest odds of sexual violence victimization whilst studying aboard.
Danehorn, 2023[40]	Cross-sectional	May to Nov 2018	Alcohol and recreational drugs including cannabis	N/A	Exchange students ($n = 48$)	N/A	Exchange students increased weekly consumption of alcohol semester abroad.	N/A

(*Continued*)

Table 1 Continued

Author, year	Study design	Duration of study	Type of substances studied	Travel destinations	Type(s) of traveller (n)	Prevalence of substance abuse	Pattern/characteristics of substance abuse	Problems from substance abuse[a]
Friedman, 2023[32]	Qualitative	2021–2022	Fentanyl, heroin and methamphetamine	Mexico	International travellers (n = N/A)	N/A	Travellers can buy fentanyl, heroin and methamphetamine-based counterfeit pills without prescription as single pills or in bottles from pharmacies located in tourist-serving areas. Advertisements of these medications usually involve erectile dysfunction medications and painkillers.	N/A
Lerksuthirat, 2023[30]	Qualitative	May to Jun 2022	Cannabis	Thailand	Local and international travellers (n = N/A)	N/A	Twitter search revealed that topic related to cannabis tourism was exclusively surfaced in English twitters. The topics had oriented towards boosting agricultural and tourism sectors with cannabis tourism after the legalization of cannabis in Thailand.	N/A

[a]Including both health- and non-health-related problems. [b]Including all travellers, not specific to drug users. [c]Including both locals and travellers. [d]These numbers are prevalence in students who reside with friends, live at home alone or live in student dormitory (which international students are part of), but these numbers did not specify the exact prevalence among international students.

Table 2 Characteristics of common recreational substances used among travellers

Substance	Estimated prevalence[a]	Popular tourist destination	Travellers' characteristics	References
Cannabis	9.5–34.5%	Netherlands, US, Canada, Denmark, Spain, Portugal, Costa Rica, Jamaica, Czechia, India, Israel, Morocco and Uruguay	Young, long-term travellers, backpackers, occupational travellers, spiritual travellers and seeking local authentic experiences	Beny et al.,[41] Bonny-Noach et al.,[27] Hughes et al.,[7] Klunge-de Luze et al.,[9] Paz et al.,[21] Pereira et al.,[2] Segev et al.[11]
Alcohol intoxication	20.4–35.9%.	Worldwide, but particularly in clubbing, dancing and festival environments	Young, male gender, backpackers, vacation-leisure travellers and occupational travellers	Bellis et al.,[15] Cabada et al.,[22] Hughes et al.,[7] Klunge-de Luze et al.[9]
Cocaine	2–22.2%	South America and Balearic Islands	Young, occupational travellers, backpackers	Bellis et al.,[15] Bellis et al.,[14] Hughes et al.,[17] Hughes et al.,[16] Paz et al.[21]
MDMA	2.8–40.5%	Southeast Asia and Balearic Islands	Young, backpackers and occupational travellers	Bellis et al.,[15] Bellis et al.,[14] Flaherty, et al.,[3] Hughes et al.,[17] Hughes et al.,[16] Kelly et al.,[34] Paz et al.,[21] Pereira et al.[2]
Methamphetamine and amphetamine	2–7.4%	Southeast Asia and Balearic Islands	Young, backpackers and occupational travellers	Bellis et al.,[15] Bellis et al.,[14] Flaherty, et al.,[3] Hughes et al.,[17] Hughes et al.,[16] Pereira et al.[2]
Psychedelic agents	2–15%	Latin America and Southeast Asia	Spiritual travellers, seeking local authentic experiences and seeking alternative medical treatment	Belhassen et al.,[35] Bellis et al.,[15] Flaherty, et al.,[3] Paz et al.,[21] Pereira et al.,[2] Prayag et al.,[23] Segev et al.[11]

[a]Prevalence among travellers who use recreational substances.

and MDMA.[3,7,21] A summary of various substances used among travellers is presented in Table 2.

Cannabis

Between 9.5% and 34.5% of travellers used cannabis and its derivatives (marijuana, hashish, etc.).[14,15,17] Cannabis stands out as one of the most commonly abused substances, constituting 69.5–87% of the cases.[7,9,11,21] The high prevalence of cannabis abuse among travellers could be linked to the legalization or decriminalization of cannabis for tourism purposes.

At-risk alcohol consumption

Approximately 14% of travellers engaged in at-risk consumption [more than eight standard drinks (SD) per week for females and 15 SD for males].[9] The reported prevalence of alcohol intoxication whilst travelling abroad falls within the range of 20.4–35.9%.[7,22]

Cocaine

Cocaine use among travellers range between 2% and 22.2%.[14–17,21] Cocaine accounted for 18.3% of the recreational drugs used by travellers.[7] It is one of the most used hard drugs among international travellers.[3,7,21]

MDMA (ecstasy and molly)

MDMA use by travellers ranging from 2.8% to 40.5%.[14–17,21] MDMA was found to be the second most popular drug used by travellers after cannabis, accounting for 31.9% of drugs used by travellers.[7]

Methamphetamine and amphetamine

Between 1.2% and 7.4% of travellers used amphetamine,[14–17] whilst 2% used methamphetamine. [15] Amphetamine accounted for 5.7% of the recreational drugs consumed by travellers.[7]

Psychedelic agents

The usage of psychedelic agents, such as LSD, ayahuasca, magic mushroom and *San Pedro* cactus, among travellers is ~2–15%.[11,15,21] The prevalence is particularly high among those travelling to tropical regions, such as Latin America and Southeast Asia.[3,23–26]

Other substances

Other recreational substances, such as GHB, ketamine and heroin, exhibit comparatively lower usage rates among travellers.[21] Reported prevalence rates for GHB, ketamine and heroin use range from 0.5% to 3.5%, 0.9% to 7.1% and 1%, respectively.[14–17]

Contributing factors to substance abuse

Several characteristics influence the drug abuse patterns of international travellers. These characteristics may include age, gender, pre-existing substance usage, travelling preferences and destinations.

Demographics of drug abuse among travellers

Several studies have highlighted that individuals in the younger age group, particularly those aged 16–35, exhibit a higher tendency to engage in substance use and alcohol consumption whilst travelling,[7–9,16,17,21] primarily by engagement in nightlife activities.[15] Consequently, resorts in popular destinations catering to youthful travellers often emphasize nightlife offerings that might involve the availability of substances and drugs.[16]

Substance abuse is more prevalent among male travellers than females, with gender-related substance abuse influenced by travel destination and style, such as backpacking.[9,15,16,21,22]

Existing substance use patterns

Numerous studies have indicated that travellers who consume alcohol at home tend to engage in excessive alcohol and drug use abroad.[9,14] However, some studies noted the reduced drug use during travel, influenced by unfamiliarity with drug sources, regulations and health concerns.[9] Conversely, studies conducted at departure points, such as Ibiza airport, revealed the increased substance use frequency among those previously using at home.[14] It was found that up to 5% of those who refrained from drug use at home resumed such behaviour whilst travelling.[14]

In the case of backpackers visiting Australia, holiday alcohol and cannabis use increased compared to their use at home. However, the usage of ecstasy, cocaine, amphetamine, methamphetamine, LSD, ketamine and GHB was lower, which could be attributed to length of their stay overseas, financial constraints, work obligations and challenges in sourcing drugs.[15] Notably, in Ibiza, travellers exhibited reduced usage frequencies of substances like amphetamine, ketamine, cannabis, LSD, cocaine and GHB compared to their home environments due to a constrained availability of less common drugs.[13] However, upon acquiring these substances, the frequency of use saw a substantial surge.[13]

Travel destination

In the context of drug tourism, certain travellers deliberately choose specific destinations to gain access to drugs. Destinations known for their vibrant nightlife and festivals might also draw in drug tourists.[3,21] Holiday resorts, in particular, may foster environments that encourage high-risk alcohol consumption without sufficient consideration for travellers' health.[7]

European destinations have exhibited a notable prevalence of substance abuse.[7,9,14] The Netherlands, Denmark, Czechia and Portugal have licenced or drug-friendly shops.[2,3,27] Some destinations decriminalized cannabis to attract tourists, such as Bermuda, Jamaica, Canada, selected US states and Thailand, which offer cannabis in shops, restaurants and nightclubs.[3,27–29] Cannabis usage trends and legal status are depicted in Figure 2.

Asia and Latin America are increasingly emerging as popular regions for drug tourism, with a considerable number of drug abuse incidents recorded among tourists in countries such as India, Thailand, Peru, Bolivia and Brazil.[21] Latin America is notably associated with the consumption of psychedelic agents.[23–25] Other popular destinations where recreational use of psychedelic mushrooms occurs include Bali, Indonesia and Koh Samui and Koh Pha-ngan islands in Thailand.[3,26]

Music festivals often facilitate drug access for travellers, a trend evident in multiple music festivals across Asia, Europe and the US.[2,4,31] Various substances, including opioids, MDMA, hallucinogenic mushrooms, marijuana and methamphetamine, attract travellers to the Golden Triangle in Southeast Asia.[2,3] Tourists are also drawn to Thailand from its full moon and floating parties.[2,3] Notably, Bolivia and Colombia host the cocaine bar, offering easy access to cocaine.[2,4] Uruguay's marijuana legalization is only limited to citizens, yet illegal markets serve drug tourists from other South American countries.[2] In Mexico, travellers can buy ketamine, heroin and methamphetamine-based counterfeit drugs as pills or bottles from pharmacies located in popular tourist areas without medical prescription.[32]

Specific destinations, such as the Balearic Islands, report substance abuse rates exceeding 80%, particularly among young and occupational travellers.[7,14,16,17] Ibiza and Majorca, known for nightlife, music events, climate and affordability, are the most popular for parties and nightlife which attract youth for substance abuse.[3,7,16] For a comprehensive overview, Supplementary File 3 highlights the most prominent drug tourism destinations worldwide.

Purpose of travel and type of travellers

Vacation and leisure trips are often associated with risky alcohol consumption, particularly among young travellers.[9,15,33] Occupational travellers often enjoy complimentary nightclub entry and discounted beverages which is reflected in high rate of substance abuse among young travellers seeking part-time employment in Ibiza.[17] Their drug abuse prevalence ranged from 13.6% to 68.9%, with ecstasy and cocaine being the most common used substances.[34] Rates of new drug users within this group ranged from 7.7% to 16.5%. Newly arrived occupational travellers often assimilated into the drug scene, influenced by peer familiar with drug sources.[34]

Certain spiritual travellers intentionally seek to disconnect from modern daily life, driven by curiosity, a quest for spiritual realization, a desire for alternative treatment and an engagement in genuine local experiences using hallucinogenic agents.[2,3,23,35,36] Examples include *kif* in Morocco, *bhang* in India and ayahuasca and *San Pedro* cactus in South America.[3,23]

Long-term travellers and backpackers show higher substance abuse rates.[15,21] The duration of travel correlates with the use of more potent drugs.[11] Young adult backpackers, seeking 'time-off' or a summer break, may adopt substances due to loneliness, cultural detachment and absence of restriction.

Studying aboard can contribute to substance use in international college students.[37–40] The prevalence of cannabis use can reach up to 36%, with even higher rates observed in students studying in South Africa, Spain, Germany, Costa Rica and Australia.[39] Living with friends, with reduced parental control, can amplify urge to use substances from peer effects.[37] Poor psychological adjustment abroad can push substance use as a coping mechanism.[37]

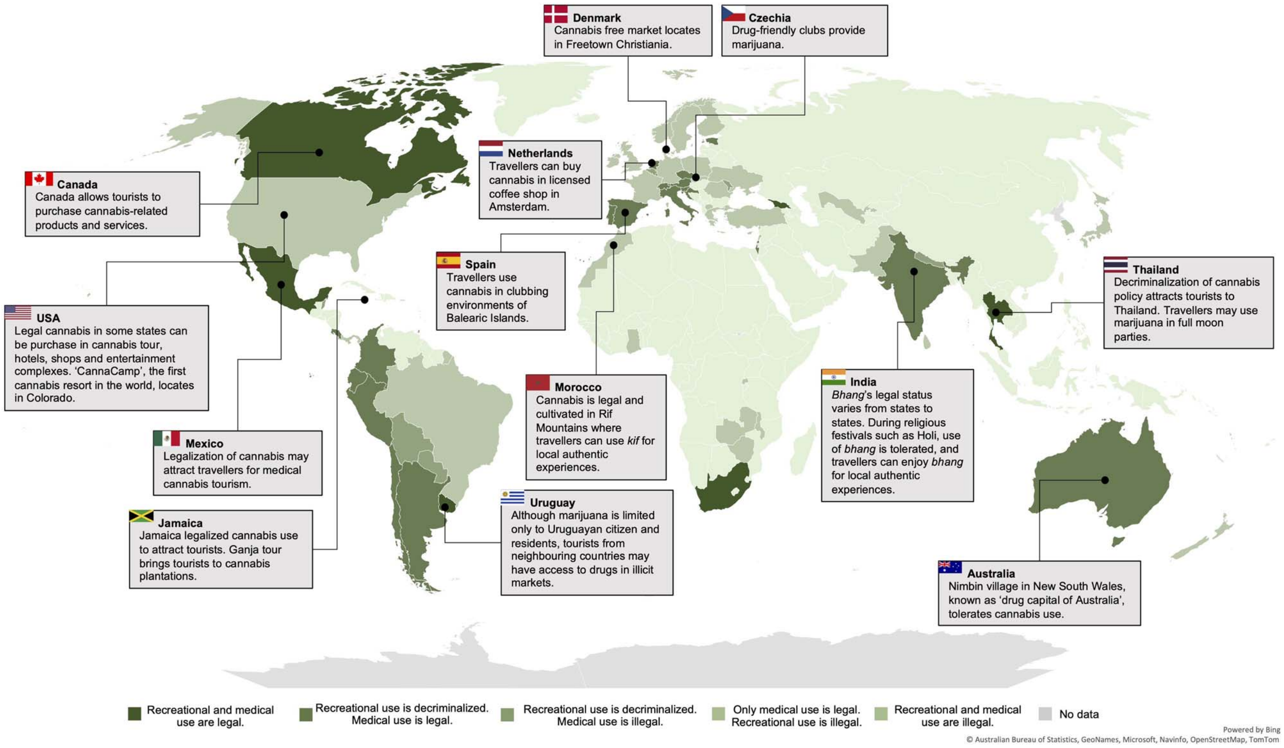

Figure 2 Legal status of cannabis in each country and pattern of use among travellers (as of August 2023)

Effects of substance abuse on travellers' health

Substances abuse by travellers can lead to significant health consequences, including violence, accidents and engaging in unprotected sexual activities that may require medical attention and potentially disrupt travel plans. The effects of substance abuse can be categorized into direct and indirect health problems.

Direct effects of substance abuse

Direct health problems arising from substance abuse are the result of the pharmacologic effects of the substances. These problems can be classified into two groups: neuropsychiatric effects and physical effects. The summary of clinical symptoms and primary management of substance intoxication are summarized in Table 3.

Neuropsychiatric effects. Recreational drugs abuse accounted for 22% of returning travellers exhibiting neuropsychiatric symptoms.[10] These neuropsychiatric symptoms from recreational drugs use included panic, depressive disorders, anxiety disorders and acute psychosis.[41]

In some instances, the utilization of recreational substances can worsen underlying psychological disorders.[42] When combined with time-zone change, substance use might lead to depression-like affective symptoms among some travellers.[12] Alcohol consumption can disrupt the circadian rhythm and aggravate jet lag symptoms.[8] Benzodiazepines, used by travellers to alleviate jet lag, can lead to 'traveller's amnesia' when mixed with alcohol, which was rare and limited to only a case report.[8]

Psychedelics may lead psychoactive effects.[43] It can also lead to 'bad trips' which may be the complications of the more severe serotonin syndrome.[43] The common adverse effects associated with substance abuse are demonstrated in Table 3.

Physical effects. Limited available literature details the prevalence and common physical health effects from substance abuse in travellers. As most travellers opt for 'soft drugs' rather than 'hard drugs' during their journeys, the resulting physical adverse effects might not be as severe.

Travellers with chronic respiratory conditions should avoid in-flight alcohol.[8] Due to the hypoxic cabin conditions, together with alcohol's respiratory effects, hypoxic conditions of the travellers could be worsened.[8]

Several fatalities have been reported among travellers in connection with ayahuasca.[25,43] Ayahuasca contains monoamine oxidase inhibitors that, when combined with selective serotonin uptake inhibitors present in various antidepressants, can lead to a potentially fatal condition known as serotonin syndrome.[43]

Indirect effects of substance abuse. Indirect health problems resulting from substance abuse do not directly stem from the pharmacologic effects of the substances. Instead, they arise as consequences of the pharmacologic effects, leading to depressed cognitive functions, increased aggressive behaviours and decreased awareness of health and safety. These issues can be classified as accidents and injuries, crime and violence, increased risky sexual behaviours, sexual violence and blood-borne infections.

Table 3 Recreational substances used among travellers, related clinical symptoms and primary management[64,72]

Substance	Mode of use	Common signs and symptoms of intoxication	Adverse health effects	Primary management of intoxicated cases
Cannabis	Smoking, vaporizing, mixing with food	Hallucination, CNS depression and miosis. Severe intoxication may have respiratory depression, hypothermia and bradycardia.	Impaired memory, psychomotor skill and attention, panic, anxiety disorder and depression	Respiratory support if indicated, naloxone as antidote for opioid intoxication and supportive care
Alcohol intoxication	Drinking	Hallucination, CNS depression, aggression	Impaired memory, psychomotor skill and attention, panic, anxiety disorder and depression	Respiratory support if indicated and supportive care
Cocaine	Snorting, rub into gum, injection, smoking	Agitation, aggression, euphoria, hyperarousal state, mydriasis, diaphoresis, tachycardia, hypertension, hyperthermia	Feeling of euphoria, hypersensitivity and paranoia. Long-term use effects include cognitive impairment, respiratory infection, anosmia and cardiovascular collapse. Overdose can lead to seizure, rhabdomyolysis, heat stroke, hypertensive crisis, myocardial infarction and cardiac arrest.	Protect airway, cooling, benzodiazepine, hydration, obtain ECG for evaluation and supportive care
MDMA	Ingesting capsule/tablet, swallowing liquid form, snorting	Euphoria, hyperarousal state, energetic, emotional closeness, diaphoresis, mydriasis, tachycardia, hyperthermia and hypertension	Seizure, rhabdomyolysis, heat stroke, hypertensive crisis, myocardial infarction and cardiac arrest.	Protect airway, benzodiazepine, cooling, activated charcoal may be used if ingested < 1 h and supportive care
Methamphetamine and amphetamine	Smoking, ingesting pills, snorting, injection	Agitation, aggression, euphoria, hyperarousal state, mydriasis, diaphoresis, tachycardia, hypertension, hyperthermia	Emotional and cognitive problems, mental health problems, weight loss, sleep problems and violence behaviour.	Protect airway, cooling, benzodiazepine, hydration, obtain ECG for evaluation and supportive care
Psychedelic agents	Ingesting raw, boiled or cooked with other food	Euphoria, hyperarousal state, hallucination, delusion, CNS stimulation, CNS depression, seizure, cholinergic effects and gastroenteritis	Feeling euphoria, relaxation, delusion, altered perception, hallucination, anxiety, paranoia and 'bad trip' (agitated, confused, disoriented, anxious and acute psychotic episode).	Benzodiazepine for treating stimulation symptoms and seizure, activated charcoal, atropine as antidote for cholinergic symptoms and supportive care

CNS, central nervous system.

Accident and unintentional injuries. Substance abuse can impair cognitive functions, increasing the risk of accidents. Alcohol significantly contributes to traffic accidents in travellers.[44,45] A study in the Balearic Islands revealed that alcohol and recreational psychoactive drug consumption led to unintentional injuries in ~6–15% of travellers.[7] Similarly, a study in Thailand found that 13.2% of traffic accident cases in travellers consumed alcohol.[46] Among hospitalized and fatal cases of travellers from traffic accident, 77.5% and 95.3% might have used recreational substances, respectively.[46] Other alcohol-related accidents in travellers encompass incidents such as drowning, falls and attacked by wild animals.[45,47,48]

Crime and violence. Among travellers, the incidence of engagement in violent activities due to substance abuse is estimated to be ~4–15%.[7] Prolonged periods of intoxication are linked to higher rates of violent behaviour.[7] A study identified frequent episodes of drunkenness and the consumption of cannabis or cocaine as indicators of the likelihood of getting involved in fights whilst on vacation.[16]

Risky sexual behaviours and sexual violence. Substance abuse can lead to unintended sexual behaviours whilst travelling.[49,50] Several studies have highlighted the association between substance abuse and engaging in risky sexual activities, such as having multiple sexual partners and lack of condom usage.[22,49,51] Pre-sex alcohol consumption has also been linked to having multiple sexual partners, whilst drug use is associated with a lack of condom usage.[49]

Use of a substance during sexual activities, dubbed as 'chemsex', is as high as 42.7% among men who have sex with men (MSM) whilst travelling in western countries.[20] Among the MSM chemsex travellers, the prevalence of STD is as high as 26.7%.[20] A study in South Florida demonstrated a strong correlation between alcohol consumption, club drug use and unprotected sexual activities among MSM travellers.[18] Substances like ketamine and 'poppers' were frequently used.[18,19]

Although not extensively studied among travellers, lone female travellers face the potential danger of sexual violence resulting from drink spiking.[52] Alcohol is frequently utilized in these incidents, whilst recreational drug use accounts for an average of 15%.[52]

Blood-borne infections. Approximately 10–20% of travellers engage in behaviours that put them at risk of acquiring HIV, including sharing of needles, syringes or equipment, during their journeys.[53] Whilst this infection is mainly associated with unprotected sexual activity, instances of drug injection have been reported among travellers.[54,55] A study from Denmark revealed that 17% of travellers engaged in drug injections whilst visiting areas with a high prevalence of hepatitis B.[55] Additionally, another study indicated that shared injections accounted for 3% of Australian travellers who were at risk of contracting blood-borne infections.[54]

Body packing complications. Body packing involves the ingestion of drugs concealed within latex sheaths or plastic bags to avoid detection whilst crossing international borders. This practice extends beyond the gastrointestinal tract to include sites, such as the mouth, vagina, ear and even the penile foreskin.[56]

Complications arising from body packing are collectively referred to as 'body packer syndrome', encompassing a range of issues such as chemical complications from drug absorption and toxicity, as well as mechanical complications, such as gastrointestinal obstruction.[56] Leaking packages cause sympathomimetic effects, leading to aggression, emergencies and in-flight death.[3] Chemical intoxication mortality reached 56% in the past, but improved packing lowered it to under 3%.[56] Mortality rates attributed to bowel obstruction from body packing are ∼5%.[56]

Non-health-related problems from substance abuse among international travellers

In addition to the health consequences for travellers, substances abuse can lead to various non-health-related problems, such as air rage and violation of local laws at their destination. The potential consequences of these issues encompass deportation, physical harm, imprisonment and in extreme cases, even facing the death penalty.

Air rage and deportation

Alcohol consumption and the use of recreational or therapeutic drugs rank among the primary contributing factors to air rage.[57,58] The preemptive use of certain prescribed drugs, such as benzodiazepines for mitigating jet lag, could potentially trigger air rage incidents, although the evidence for this relation is not conclusively established.[57,58] No direct evidence supports the notion that using recreational drugs before boarding a flight causes air rage.

Violation of local laws

Different countries and territories have varying regulations concerning substance use which may confused travellers.[59] Typically, alcohol is obtainable subject to specific regulations in certain destinations, whilst cannabis is subject to various restrictions.[27,28] Conversely, trafficking, importing and exporting of hard drugs are universally illegal. Numerous foreign nationals face severe penalties for drug-related offences in multiple countries, including life imprisonment and death sentences.[8] In certain instances, substance-using tourists might be required to enrol in lengthy rehabilitation programmes.[8,26] The penalties for drug-related crimes in each country are demonstrated in Supplementary File 4.

Legally prescribed substances might face customs issues upon arrival.[8,60] Whilst substances, such as tetrahydrocannabinol, might be legally permissible for medicinal use in a travellers' departure countries,[61] they could be prohibited upon arrival in their destination country, and attempting to carry them across borders could result in penalties.[60]

Discussion

Substance abuse among travellers is an underestimated problem, exacerbated by carefree and uninhibited behaviours often observed during holiday breaks, providing an escape from responsibilities and domestic constraints.[62,63] The prevalence of substance abuse among travellers surpass the overall global prevalence, which ranges from 0.05 to 14.53%.[1] Cannabis emerges as the most commonly used substance among both travellers and locals,[1,7,8,11,21] potentially linked to its legalization.[27,28]

In destinations renowned for offering recreational substances, such as the Balearic Islands, the prevalence of substance abuse can be alarmingly high.[7,14–17] This could be attributed to international travel by a specific group of individuals seeking destinations known for providing recreational substances. It is crucial to note that these numbers may include travellers who consume small amount without reaching intoxication level, potentially leading to a misleading elevation of the overall prevalence of substance abuse among travellers. Furthermore, numerous studies focus on specific traveller groups more prone to substance abuse, such as young travellers, backpackers and long-term travellers.[15–17,21,32]

The prevalence of recreational substance abuse, including cocaine, MDMA, methamphetamine and amphetamine, may be either under- or overestimated due to concerns about legal jurisdiction, recall bias and confusion with other types of drugs.[8,15]

Whilst short-term engagement in recreational drug abuse can foster dependence and relapse.[14] The long-term effects of dependence can result in various mental health issues, such as cognitive impairment, emotional disturbances, memory lapses and aggressive behaviours.[64] Several cases of travellers afflicted with mental health issues from substance abuse require emergency visits, and even prolonged treatment and rehabilitation.[41] In certain travel destinations, the unpredictability of medical treatment cost, duration and quality is noteworthy. It is important to emphasize that travel insurance policies often do not cover the medical expenses associated with drug-related issues.[8]

Estimating the precise prevalence of physical effects resulting from the direct pharmacologic effects of substance abuse is challenging due to limited available literature. Since most travellers tend to opt for 'soft drugs' rather than 'hard drugs' during their journeys,[7,9,11,21,22] the resulting physical adverse effects may not be as severe and could be underreported.[7] Nevertheless, excessive abuse can directly lead to adverse physical effects, particularly when substances are consumed in substantial quantities. Stimulants, such as cocaine, amphetamine, methamphetamine and MDMA, heighten metabolism through dopaminergic pathways, raising blood pressure, cardiovascular workload, dysrhythmias and life-threatening hyperthermia.[59,65,66]

The links between substance abuse and indirect health problems are complex due to several factors.[7] Environments where travellers have access to substances may already feature violence and crime, such as drug markets and nightclubs.[7] Additionally, certain groups of travellers, such as MSM travellers, may enhance their travel experiences by using substance as a complementary factor, such as engaging in substances use during sexual activities.[19,20,67] Thus, the prevalence of indirect health problems resulting from substance abuse can be substantial.

Travellers should familiarize themselves with local regulations regarding recreational substances in their destination countries. Blood alcohol limits vary across nations and might be lower than those in the traveller's home country.[68,69] Some countries restrict alcohol with age limits, purchase times and alcohol-free zones.[69] Most Islamic countries have implemented a complete ban on alcoholic beverages.[69,70] Certain narcotics and psychotropic medications, such as ketamine, serve medical purposes.[71] Travellers with conditions such as attention deficit hyperactivity disorder or narcolepsy who carry prescribed amphetamines may face arrest upon entry.

Travel health practitioners can utilize data on the prevalence and destination-specific patterns of substance abuse, traveller profiles, potential harms and local regulations. By integrating these factors, practitioners can provide advice to travellers with itineraries that might heighten the risk of substance abuse.

Since travellers visiting travel clinics are generally more aware of health risks and may not be the target for pre-travel advice on the risk of substance abuse,[9] disseminating educational messages about substance-related health risks, regulations and available local medical assistance to the true target travellers can effectively mitigate harm.[13] These messages should target various stakeholders such as travel agencies, airlines, hotels, ports of entry and tourist information centers, especially those operating in drug tourism destinations. With this approach, it will be possible to expand health education beyond dedicated travel clinics to a wider range of travellers.[13]

In addition to general travellers, travel health practitioners should advise travellers prescribed with narcotic and psychotopic medications to obtain a certificate from their departure authorities, validating their legal drug possession and usage upon entry. The prescription should come from the treating physician, and a copy should be carried by the traveller.[71] Recommended drug quantities should align with a 30-day treatment, as per International Narcotics Control Board guidelines.[60]

Healthcare providers may encounter syndromes arising from recreational substance abuse during and after travel, presenting as panic attacks, anxiety, depression, aggressive behaviours and even coma.[10,41] These symptoms may overlap with those linked to mefloquine chemoprophylaxis, physical fatigue or underlying psychiatric issues.[10] In severe cases overseas, repatriation might be necessary.[41] Whilst it is rare for a traveller with syndromes resulting from substances abuse to present at travel clinics, as most will seek emergency rooms or outpatient departments, it should be noted that a travel history involving drug tourism destinations and substances used whilst travelling can be helpful for making differential diagnosis and treatment for physicians encountering travellers with these symptoms.

This review has notable limitations. The majority of studies were conducted on European travellers journeying within Europe, with a particular focus on the Balearic Islands. Few studies explored travellers to other destinations where drug tourism is prevalent. Studies in these drug tourism destinations may be subjected to population selection bias, potentially leading to an overestimated prevalence. Moreover, several studies had small sample sizes, which may not accurately represent the general population of travellers. The inclusion of neuropsychiatric problems in studies was often not classified as resulting solely from recreational substance abuse but was rather attributed to various factors, including malaria chemoprophylaxis or environmental influences.[41] This hampers an accurate portrayal of neuropsychiatric issues specifically linked to recreational substance abuse among travellers.

Conducting epidemiological research on the prevalence and risk factors of substance abuse among a broader spectrum of travellers to different destinations, other than Europe, is crucial for understanding the pattern of substance abuse among general travellers. Descriptive and epidemiological studies that address the true prevalence and the risk factors of substance-related problems, whether health-related or non-health-related, can help identify travellers with these risks who should be the target of health education messages. Lastly, the intervention studies, such as those addressing pre- and post-intervention awareness about substance abuse in travel destinations or those comparing the substance-related problems in intervention and control groups, are important for developing tools and approaches to prevent travellers' health issues arising from recreational substance abuse.

Conclusion

The rising concern of substance abuse poses a significant threat to international travel, particularly impacting young, long-term and adventurous backpack travellers. Several nations have embraced lenient drug policies, aiming to curb illicit drug use whilst attracting tourists. The dangers of substance abuse among travellers cast a foreboding shadow, capable of causing harm on health, journey, serenity and security. In this high-stakes landscape, it is crucial for travel health practitioners, together with various stakeholders, to provide essential interventions that extend beyond dedicate travel clinics. These interventions should address the health risks, holistic well-being and the intricate local regulations concerning substance abuse, especially in destinations entwined with the allure of drug tourism.

Supplementary data

Supplementary data are available at *JTM* online.

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Author contributions

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